What Is a Topos? From Local Data to a Universe of Variable Sets

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Imagine measuring a physical field over a region. One instrument knows what happens on the left, another on the right, and neither has the whole picture. If they agree wherever their coverage overlaps, can we regard their results as one measurement? That is natural for continuous functions, but the same question becomes more subtle for solutions of equations, sections of coverings, or the truth of statements about a changing system.

Sheaf theory organizes local observations into global objects. Topos theory goes further: it finds categories of local data in which we can perform many of the constructions ordinarily performed with sets, functions and logical predicates. The surprise is that truth can depend on an open region, and some classical logical principles need not hold internally. On an open region, a claim may hold locally even when it does not hold globally; restriction maps determine precisely how such local truths fit together.[1]

§1.1 The Locality of Sheaf Elements

Let X be a topological space and $U \subseteq X$ open. The continuous real-valued functions on U form a set

$$\mathcal{C}(U) = \{f : U \to \mathbb{R} : f \text{ is continuous}\}.$$

If $V \subseteq U$, we can restrict a function:

$$\rho_V^U : \mathcal{C}(U) \to \mathcal{C}(V), \qquad f \mapsto f|_V.$$

Restriction obeys two laws:

$$\rho_U^U = \mathrm{id}, \qquad \rho_W^V \circ \rho_V^U = \rho_W^U \quad (W \subseteq V \subseteq U).$$

Keeping only $\mathcal{C}(X)$ would discard local functions that cannot be extended to all of X. We therefore regard the family of sets $\mathcal{C}(U)$, together with its restriction maps, as a single object.

A *presheaf* is an assignment $U \mapsto F(U)$ with exactly these restriction laws. Formally, open sets form a category with an arrow $V \to U$ whenever $V \subseteq U$; the direction of the restriction maps is reversed:

$$\boxed{F : \mathrm{Open}(X)^{\mathrm{op}} \longrightarrow \mathbf{Set}.}$$

An element of $F(U)$ is a *section over* U. It need not be an ordinary function. It can represent a solution to an equation, a choice of coordinates, or a possible state of a system on that region.

[1] For these constructions see Mac Lane and Moerdijk, *Sheaves in Geometry and Logic*, and Fong and Spivak, *Seven Sketches in Compositionality*, Chapter 7.

§1.2 Compatible local descriptions should glue

Suppose $U = U_1 \cup U_2$. Two continuous functions $f_1 \in \mathcal{C}(U_1)$ and $f_2 \in \mathcal{C}(U_2)$ satisfying $f_1 = f_2$ on $U_1 \cap U_2$ define

$$f(x) = \begin{cases} f_1(x), & x \in U_1, \\ f_2(x), & x \in U_2. \end{cases}$$

The overlap condition makes the definition independent of which local formula we choose. It is continuous, and it is the unique function whose restrictions are f_1, f_2.

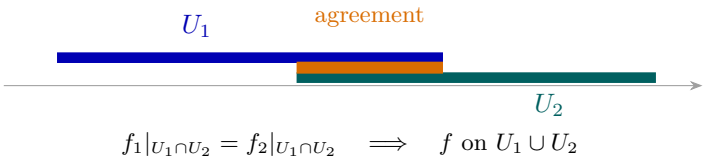


Figure 1.1: The equality on the overlap is precisely what makes one global function from two local functions.

A *sheaf* is a presheaf for which this construction works on every open cover $U = \bigcup_i U_i$. Whenever $s_i \in F(U_i)$ agree on all intersections $U_i \cap U_j$, there exists one and only one $s \in F(U)$ restricting to every s_i. The existence clause and the uniqueness clause are both essential.

Consider a presheaf with

$$F(U) = \{\text{red}, \text{blue}\} \quad (U \neq \varnothing), \qquad F(\varnothing) = \{*\},$$

with the obvious constant restrictions on nonempty opens. Choose two disjoint nonempty intervals U_1, U_2. Red on U_1 and blue on U_2 agree vacuously on the empty overlap, but neither of the two globally constant options in $F(U_1 \cup U_2)$ restricts to both. This presheaf is not a sheaf.

The corresponding sheaf consists of *locally constant functions* $U \to \{\text{red}, \text{blue}\}$. It permits red on one connected component and blue on another. The point of sheafification in this elementary example is to recognize combinations of local choices that the topology really allows.

§1.3 Square Roots around the Circle

Let $X = S^1 \subset \mathbb{C}$. On every open arc U, consider the set

$$\mathcal{R}(U) = \{w : U \to S^1 \text{ continuous} : w(z)^2 = z \text{ for all } z \in U\}.$$

These local roots restrict to smaller arcs and glue uniquely whenever they agree. Thus $\mathcal{R}$ is a sheaf.

Locally there are two roots, but there is no global continuous root $w : S^1 \to S^1$ satisfying $w(z)^2 = z$. If it existed, the degree of circle maps would obey

$$1 = \deg(\mathrm{id}_{S^1}) = \deg(w^2) = 2\deg(w),$$

an impossibility since degrees are integers.

The obstruction is easy to visualize. Write $z = e^{it}$. A local choice is $w = e^{it/2}$. As t makes one turn from 0 to 2π, the root moves from 1 to -1. Its sign is exchanged, so the starting choice fails to close up.

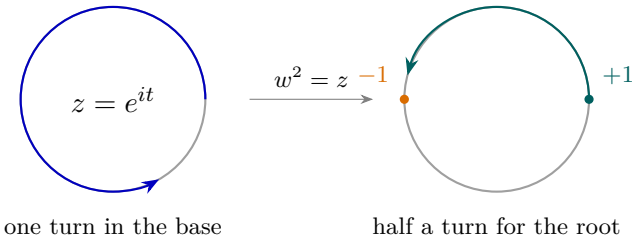


Figure 1.2: One circuit of the base exchanges the two local square roots. Local solvability does not imply the existence of a global choice.

There is an actual geometric object behind this example:

$$E = \{(z, w) \in S^1 \times S^1 : w^2 = z\}, \qquad E \longrightarrow S^1, \quad (z, w) \longmapsto z.$$

This is a two-sheeted covering. The sheaf $\mathcal{R}$ is its sheaf of local sections. Calling a sheaf a set varying over a space now has a concrete meaning: every small region has a set of available choices, but the choices can twist when transported around loops.

§1.4 From subsets to characteristic maps

In **Set**, a subset $A \subseteq S$ can be recorded as a characteristic function

$$\chi_A : S \to \{0, 1\}.$$

Let $1 = \{*\}$ and let $\text{true} : 1 \to \{0, 1\}$ select the value 1. The defining property of χ_A is expressed by the pullback square

$$\begin{array}{ccc} A & \longrightarrow & 1 \\ \downarrow & & \downarrow{\scriptstyle \text{true}} \\ S & \xrightarrow[\chi_A]{} & \{0, 1\}. \end{array}$$

A pullback is the categorical version of imposing two compatible constraints: here it selects the elements of S for which the characteristic function is true.

An injective map into S, considered up to isomorphism over S, is a *subobject*. The pair consisting of $\Omega = \{0, 1\}$ and the truth map is a *subobject classifier*: every subobject arises as the pullback of truth along a unique characteristic map.

This description uses arrows and universal properties rather than elements. So it can be repeated in a sheaf category, even though elements there are locally defined sections rather than points.

§1.5 Open Sets as Truth Values

In $\mathbf{Sh}(X)$, the subobject classifier Ω has a remarkably simple form:

$$\boxed{\Omega(U) = \{V \subseteq U : V \text{ is open in } X\}.}$$

Restrict a truth value $V \subseteq U$ to $W \subseteq U$ by $V \mapsto V \cap W$. Complete truth on U is the whole open set U; complete falsehood is $\varnothing$.

Here is why this classifier works. Given a subsheaf $G \hookrightarrow F$ and $s \in F(U)$, form the largest region on which s belongs to G:

$$\boxed{\chi_{G,U}(s) = \bigcup \{V \subseteq U \text{ open} : s|_V \in G(V)\}.}$$

The sheaf condition for G guarantees that s restricted to this union is genuinely a section of G. Restriction to a smaller open commutes with taking this truth region, so the maps define a morphism $\chi_G : F \to \Omega$. And

$$s \in G(U) \quad \Longleftrightarrow \quad \chi_{G,U}(s) = U.$$

That is exactly the pullback-of-truth condition familiar from characteristic functions of subsets.

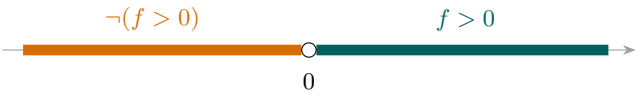
Take continuous real functions $\mathcal{C}(U)$, and let

$$\mathcal{P}(U) = \{f \in \mathcal{C}(U) : f(x) > 0 \text{ for all } x \in U\}.$$

The characteristic morphism of this subsheaf has a very concrete value:

$$\boxed{\chi_{\mathcal{P},U}(f) = \{x \in U : f(x) > 0\}.}$$

The positivity region is open because f is continuous. For the function $f(x) = x$, the predicate $f > 0$ is true on $(0, \infty)$, while its negation is true on $(-\infty, 0)$. Neither one is true throughout an open neighborhood of 0.



no open neighborhood of 0 lies wholly in either region

Figure 1.3: Truth over a sheaf is local. The boundary of a positivity region is not covered by either the predicate or its open negation.

§1.6 Why the internal logic is intuitionistic

Global truth values are elements of $\Omega(X)$, namely open subsets of X. They form a Heyting algebra:

$$\begin{aligned} U \wedge V &= U \cap V, \\ U \vee V &= U \cup V, \\ U \Rightarrow V &= \operatorname{int}((X \setminus U) \cup V), \\ \neg U &= \operatorname{int}(X \setminus U). \end{aligned}$$

The implication is the largest open W for which $W \cap U \subseteq V$. Indeed, that condition is equivalent to $W \subseteq (X \setminus U) \cup V$, so the largest open solution is its interior.

For $U = (0, \infty) \subseteq \mathbb{R}$,

$$\neg U = (-\infty, 0), \qquad U \vee \neg U = \mathbb{R} \setminus \{0\} \neq \mathbb{R}.$$

The law of excluded middle fails internally because 0 has no open neighborhood contained in either truth region.

There is an even more striking phenomenon:

$$\neg\neg U = \operatorname{int}(\overline{U}).$$

For the dense open set $U = \mathbb{R} \setminus \{0\}$,

$$\neg U = \varnothing, \qquad \neg\neg U = \mathbb{R}.$$

A proposition can be true on a dense open set without being true on the whole space, while its double negation is true everywhere. These calculations explain the intuitionistic logic of sheaves through the elementary topology of open sets. We can still reason classically about the category from outside; internal and external truth are different mathematical levels.

§1.7 Existence inside a sheaf topos is a local assertion

Return to the square-root example on S^1. Let $\mathcal{S}(U)$ be the continuous S^1-valued functions on U, and let z denote the identity function $S^1 \to S^1$. Consider the statement

$$\exists w \in \mathcal{S} : \quad w^2 = z.$$

Externally it is false if interpreted as the existence of one globally defined continuous root. Internally, however, the existential statement is true over the whole circle, because the circle has an open cover on each member of which a continuous root exists.

In the Kripke–Joyal semantics of sheaves, a statement is evaluated at an open stage U, written $U \Vdash \varphi$. The rule

$$U \Vdash \exists w\, \varphi(w)$$

means that there exists an open cover $U = \bigcup_i U_i$ and, on each U_i, a local witness w_i satisfying φ. The witnesses need not match on overlaps; if they did, the sheaf law would glue them globally.

Universal quantification is different. The assertion $U \Vdash \forall w\, \varphi(w)$ requires φ to hold for every section w over *every* smaller open $V \subseteq U$. The logical rules are constructed to respect restriction.

Here a real distinction becomes visible. Local solvability, global solvability and the existence of a globally consistent selection are three questions that should not be conflated. Topos logic lets the first be true while the third remains false.

§1.8 What Defines an Elementary Topos

An *elementary topos* is a category equipped with three types of structure.

First, it has *finite limits*: terminal objects, products, and pullbacks. These encode compatible pairs, intersections of constraints and solutions of equations.

Second, it is *cartesian closed*: for objects A, B, there exists an object B^A representing functions out of A. Precisely,

$$\boxed{\mathrm{Hom}(T \times A, B) \cong \mathrm{Hom}(T, B^A)}$$

naturally in T. This makes function spaces into objects of the same universe.

Third, it has a *subobject classifier* $1 \xrightarrow{\text{true}} \Omega$. This is the categorical mechanism for subsets and predicates.

The category **Set** has all three structures. So does the category of sheaves $\mathbf{Sh}(X)$ for any topological space X. Products of sheaves are calculated sectionwise. Their exponentials are sheaves of locally defined morphisms:

$$(G^F)(U) \cong \mathrm{Hom}_{\mathbf{Sh}(U)}(F|_U, G|_U).$$

Thus a function space in a sheaf topos is itself an object whose available functions vary from region to region.

A one-point topological space has a sheaf category equivalent to **Set**, so ordinary sets are recovered as the simplest example. An elementary topos need not satisfy excluded middle or the axiom of choice; these are additional properties. Its basic axioms also do not require a natural-numbers object, which is an extra assumption when needed.

Grothendieck topoi enlarge the geometric viewpoint: instead of open subsets of a topological space, one uses an abstract *site*, a category supplied with designated covering families. Every Grothendieck topos is an elementary topos, but not every elementary topos must arise from a sheaf construction on an ordinary topological space. The distinction prevents the attractive varying-sets picture from being overinterpreted.

§1.9 A two-stage world already has three truth values

There is a finite example requiring no topology or continuous functions. Consider the category with two objects u, v and exactly one nonidentity arrow

$$a : u \longrightarrow v.$$

A presheaf on this category consists of sets $F(v), F(u)$ and a restriction map $F(v) \to F(u)$. The stage v can be read as a broad observation and u as a more restricted one.

A *sieve* on v is a collection of arrows into v closed under precomposition. Here there are exactly three:

$$\boxed{\varnothing, \qquad \{a\}, \qquad \{\mathrm{id}_v, a\}.}$$

The middle sieve says that a property is verified after restriction along a, but not at the broad stage v. The subobject classifier of the presheaf topos $\mathbf{Set}^{\mathcal{C}^{\mathrm{op}}}$ consists of sieves stage by stage; its value $\Omega(v)$ is precisely this three-element ordered set.

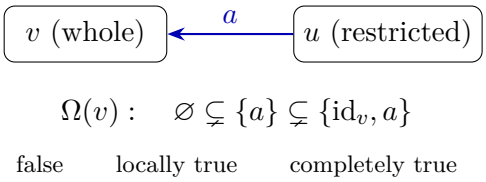


Figure 1.4: Even a category with just two stages supports a non-Boolean local truth value.

The negation of the middle sieve $\{a\}$ is empty: there is no arrow into v along which the sieve becomes empty after pullback. Thus

$$\{a\} \vee \neg\{a\} = \{a\} \neq \{\mathrm{id}_v, a\}.$$

Excluded middle fails in this finite example just as it did for opens of the real line.

For an arbitrary category $\mathcal{C}$, the subobject classifier of its presheaf category assigns to each object the collection of all sieves on that object. The open-set truth values of sheaves and the sieve truth values of presheaves express the same principle: valid information must remain valid when we pass to a more restricted viewpoint.

§1.10 Why the abstraction earns its place

Sheaves allow local models to be combined exactly when their agreement permits it. The double cover of the circle shows that local solutions can carry a global topological obstruction. The subobject classifier turns properties of those local solutions into arrows, just as characteristic functions turn subsets into arrows in ordinary set theory. And the internal logic distinguishes a global witness from a collection of local witnesses.

The same machinery is useful in several mathematical settings. A geometer may consider functions, sections and differential equations over open neighborhoods. A logician can evaluate propositions relative to contexts rather than assume every truth value is decided globally. A computer scientist can represent states or possible behaviors varying across observations or time intervals. A topos provides a common setting in which these objects, their functions and their predicates can be studied coherently.

It is thus helpful to regard a topos as a category with enough of the structural features of sets for ordinary mathematical constructions to make sense internally. The elements, however, may be locally defined, and the truth values may reflect the region or context on which they are asserted. The gluing law, the square-root obstruction and the three-element sieve show those distinctions in computations rather than slogans.